COD orders: what the data says and what I recommend

To: Founder | From: Manoj Shiva | 31 Oct 2026 | Based on 1,984 clean COD orders (Apr–Sep 2026)

The short version

Every RTO costs us **₹169** (shipping both ways, packing, damaged stock). But a delivered ₹649 order earns **₹312** after shipping and packing. So cancelling only pays when an order is more than ~65% likely to come back, and almost none are. The money is in (1) stopping unusable orders, (2) a WhatsApp confirmation for the ~14% of orders that bounce half the time, and (3) a small prepaid discount for the middle. Replaying September, this would have added about **₹5.7k on 329 sample orders (≈ ₹28k a month at ~1,650 COD orders)**, versus **–₹10.8k** if we auto-cancelled everything above 40% risk. The size of the gain rests on assumptions about how customers react, covered in section 4.

1. What surprised me

- **Cancelling risky orders loses money.** In September, cancelling every order at 40%+ risk would have cancelled 53 orders, saved ₹4.9k of RTO cost, and thrown away ₹15.8k of margin on orders that would have been delivered. Net: –₹10.8k. At 30%+ it is –₹24.6k.
- **Bad data, not new customers, is the real problem.** Unusable addresses (“xyz”, “.”, “call me”, city name only) and fake names (“Test Order”, “asdf”) are 9.5% of orders and come back **60%** of the time: 20% of all RTOs. First-time buyers do come back more (30.3% vs 23.0% for repeat) but once I set aside junk orders and the six worst cities, it is 22% vs 15%.
- **Six cities drive the rest.** Gorakhpur 62%, Bareilly 61%, Darbhanga 59%, Sagar 49%, Patna 48%, Kota 41%: 16% of orders but 29% of RTOs. Metros are 18–25%. “Stop COD in bad pincodes” can't work: 1,066 pincodes for ~2,000 orders, median one order each, none with ten. City is the smallest unit with enough data.
- **Hinglish addresses are not a red flag.** Landmark-only addresses (“Big Bazaar ke paas”) look worse overall (31%), but that is mostly because they cluster in the risky cities: 24% vs 19% for fully structured addresses in ordinary cities, 51% vs 49% in the risky ones. We use AI to understand these addresses, not to punish them.
- **Repeat refusers are the clearest signal.** Customers who refused two earlier orders: 76% RTO (34 orders). One earlier refusal: 35%.
- **The file has problems:** 30 duplicate rows, three ₹0 orders, one still in transit, 17 invalid phone numbers (47% RTO), and an average order of ₹921, not ₹649.

2. Recommended rules, applied to September (329 COD orders)

Risk of coming back	Action	Orders	Share	Actually came back
Under 20%	Ship as-is	90	27%	7.8%
20–40%	Prepaid offer (₹30–60 off)	186	57%	26.9%
40–68%	WhatsApp confirmation first	46	14%	50.0%
68%+, bad phone, junk address + fake name, 2+ earlier refusals	Hold / cancel	7	2%	85.7%

The buckets separate cleanly: the ship group returns 7.8% of orders, the confirm group 50%. Estimated money effect: RTO costs avoided ₹5.5k + sales recovered from would-be refusers ₹6.1k – margin lost on genuine orders ₹3.5k – discounts and messages ₹2.5k = **₹5.7k**, against a September RTO bill of ₹14.6k. The prepaid discount is half of break-even: about ₹40 on a ₹649 order at 30% risk (6%), ₹60 at 40% (9%); full maths in scripts/discount_math.py. All thresholds are in one settings file anyone can edit.

3. Operations vs. Marketing, in rupees

- **Operations gets** the bulk of the avoidable loss: junk and fake orders are held (7 in September, 86% of which came back), and the half-risky group has to confirm before we spend ₹130 on shipping.
- **Marketing gets** real protection: only 4 of 241 first orders were held in September; the other 203 risky first orders were confirmed or offered a discount, never cancelled. The ₹300 ad spend is already gone either way, so it is no reason to ship a bad order. What matters is the customer's future value, which I assume is ₹100 per first order (to be measured). A wrongly cancelled first order costs ~₹412; a true RTO costs ₹169, so we should only cancel when we are very sure.
- **The trade-off is one dial.** Making confirmation stricter or looser (30% to 50%) moves the September result only between ₹5.1k and ₹6.5k, so the choice is about customer friction, not money. A dashboard alarm fires if more than 8% of a day's orders are held.

4. What could go wrong, and how we'd notice

- **Customers may not react as I assumed** (reply rates, prepaid take-up). If customers respond 30% less, the net falls to ~₹0.7k. If would-be refusers never convert, about –₹1.2k (≈ –₹6k a month). Watch: WhatsApp reply rate, RTO rate within the confirm group, prepaid share. Measure these in week one.
- **We lose genuine customers quietly.** Watch: orders cancelled for silence, customers replying “YES” after cancellation, and the hold-share alarm.

- **The model is small and will drift** (1,984 orders; out-of-time AUC 0.74). Watch: predicted vs actual RTO per bucket each week; retrain monthly.
- **AI address checks can be wrong.** I tested the pipeline with the rule-based fallback; the Claude-powered check is built but not yet validated on this data. Spot-check 50 judgements before relying on it. If the AI is down, orders still flow.
- **City as a signal can be unfair to honest customers in smaller cities.** That is why those orders get confirmation or a discount, not cancellation.

5. If I had two more weeks

- **Prove it with a holdout:** leave 10% of orders untouched to measure the real effect of each intervention, and test three discount levels to replace my assumed take-up.
- Connect real WhatsApp, Shopify and payment links; build the human call queue for first orders.
- Check courier serviceability by pincode through the couriers' own APIs (Xpressbees returns 31% vs Delhivery 26%, which needs a controlled test), and add address verification and auto-correction.
- Retrain monthly and alert on drift; validate the AI address judge against a hand-labelled sample.